



20. Let EB be the line of total loss [EB = ED + DB where ED = constant loss and DB = variable loss]

21. In the line DB locate the point G to separate the stator and rotor copper losses by using the formula, $\frac{\text{Rotor Copper Loss}}{\text{Stator Copper Loss}} = \frac{I_2'^2 R_2'}{I_2'^2 R_1} = \frac{R_2'}{R_1}$ where R_1 = stator resistance per phase and R_2 = rotor resistance per phase.

$$\text{Or, } \frac{BG}{BD} = \frac{R_2'}{R_{01}} = \frac{\text{Rotor Copper Loss}}{\text{Stator Copper Loss} + \text{Rotor Copper loss}}$$

22. To get the torque line, join the points A and G

23. To find the full load quantities, draw line BK (= full load output / power scale). Now, draw line PK parallel to output line meeting the circle at point P.

24. Draw line PT parallel to Y-axis meeting output line at Q, torque line at R, constant loss line at S and X-axis at T.

Note: Choose the current scale such that the circle diagram will be as large as possible. The larger the circle diagram more will be the accuracy. Select power scale = $\sqrt{3} \times V_{\text{rated}} \times \text{current scale}$.

TABULATION

NO LOAD TEST					BLOCKED ROTOR TEST				
V _{oc}	I _{oc}	W ₁	W ₂	W _{oc}	V _{sc}	I _{sc}	W ₁	W ₂	W _{sc}

STATOR RESISTANCE MEASUREMENT			
S.No.	V (volts)	I (amps)	R _{dc} =V/I Ω
1.			
2.			
3.			
4.			
R _{dc} =			

CIRCLE DIAGRAM